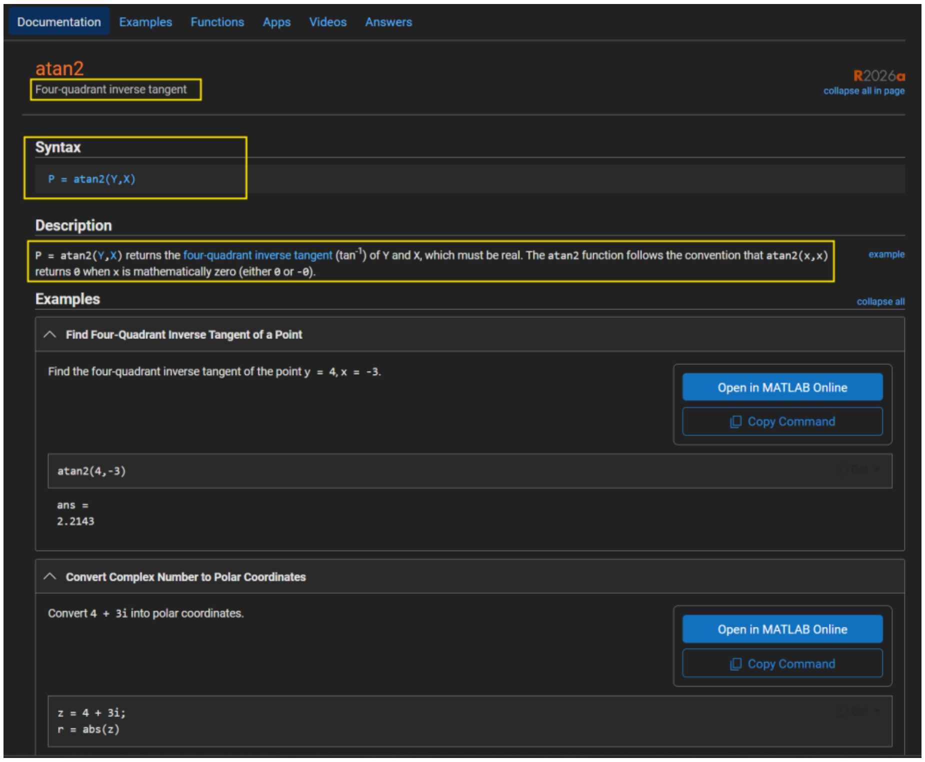
Atan2

(Arctangent Logic)

[Link to the website]



Documentation Examples Functions Apps Videos Answers

atan2

Four-quadrant inverse tangent

R2026a

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Syntax

```
P = atan2(Y,X)
```

Description

`P = atan2(Y,X)` returns the four-quadrant inverse tangent ($\tan^{-1}$) of Y and X, which must be real. The `atan2` function follows the convention that `atan2(x,x)` returns `0` when x is mathematically zero (either `0` or `-0`).

example

Examples

collapse all

Find Four-Quadrant Inverse Tangent of a Point

Find the four-quadrant inverse tangent of the point `y = 4, x = -3`.

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Copy Command

```
atan2(4,-3)
```

```
ans =
2.2143
```

Convert Complex Number to Polar Coordinates

Convert `4 + 3i` into polar coordinates.

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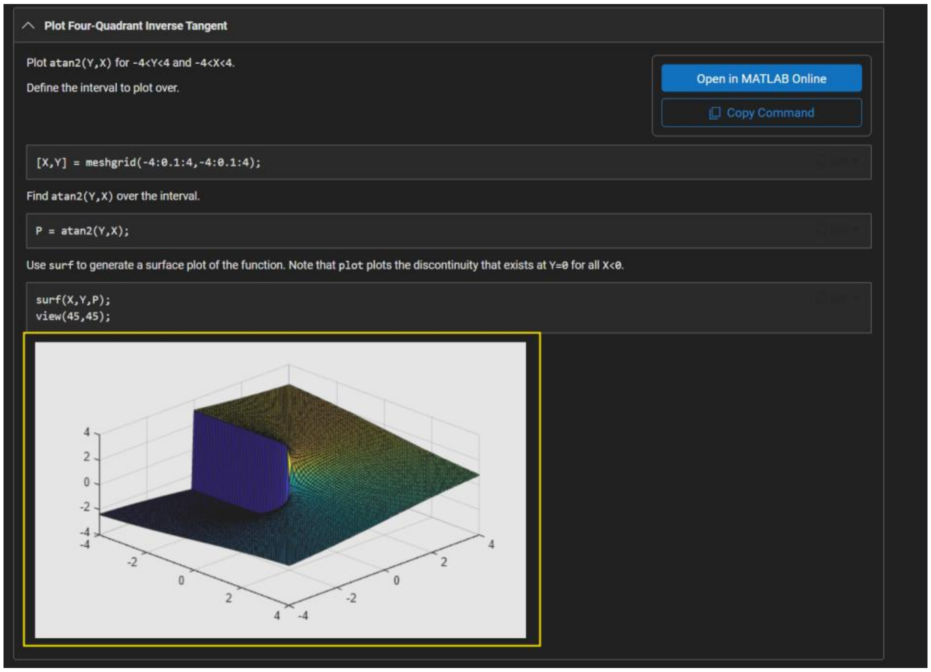
```
z = 4 + 3i;
r = abs(z)
```



Plot Four-Quadrant Inverse Tangent

Plot `atan2(Y,X)` for `-4<Y<4` and `-4<X<4`.

Define the interval to plot over.

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```
[X,Y] = meshgrid(-4:0.1:4,-4:0.1:4);
```

Find `atan2(Y,X)` over the interval.

```
P = atan2(Y,X);
```

Use `surf` to generate a surface plot of the function. Note that `plot` plots the discontinuity that exists at `Y=0` for all `X<0`.

```
surf(X,Y,P);
view(45,45);
```